Hideyuki Kawashima

Keio University
Faculty of Environment and Information Studies
Graduate School of Media and Governance
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Education

Ph.D., Engineering (Open and Environmental Systems) — March 2005
Keio University, Graduate School of Science and Technology — *Yokohama, Kanagawa, Japan*

B.Eng., Electrical Engineering — March 1999
Keio University, Faculty of Science and Technology — *Yokohama, Kanagawa, Japan*

Appointments

Visiting Professor — 2026 – present
National Institute of Informatics — *Tokyo, Japan*

Professor — 2025 – present
Keio University, Faculty of Environment and Information Studies — *Fujisawa, Kanagawa, Japan*

Associate Professor — 2018 – 2025
Keio University, Faculty of Environment and Information Studies — *Fujisawa, Kanagawa, Japan*

Associate Professor — 2016 – 2018
University of Tsukuba, Center for Computational Sciences — *Tsukuba, Ibaraki, Japan*

Lecturer — 2007 – 2016
University of Tsukuba, Graduate School of Systems and Information Engineering — *Tsukuba, Ibaraki, Japan*

Research Associate — 2005 – 2007
Keio University, Faculty of Science and Technology — *Yokohama, Kanagawa, Japan*

Research Interests

Database management system kernels; concurrency control and transaction processing; logging and recovery; distributed and replicated data systems; data stream processing; data platforms for robotics and artificial intelligence; quantum computing for data management problems.

Teaching Experience

- Algorithm Science (Keio University, SFC) – AY2026
- Big Data System (Keio University, SFC) – AY2026

- Special Lecture on Data Science A/B (Keio University, SFC) – AY2026
- Advanced Research (Cyber Informatics) (Keio University, SFC) – AY2026
- Academic Project: Novel Computing – AI and Brain Science (Keio University, SFC) – AY2026
- Master's Research Seminar (Keio University, Graduate School of Media and Governance) – AY2026

Awards & Honors

- Outstanding Research Award, xSIG, 2026 – "Phantom Arena: Decoupling Phantom Avoidance from Index Structure"
- Outstanding Paper Award, CANDAR, 2024 – "Serval: A Wait-free Multi-version Deterministic Concurrency Control Scheme"
- Best Poster Award, IEEE CLUSTER, 2024 – "Cheetah: An Efficient Deterministic Concurrency Control Scheme with Non-Visible Write Elimination and Re-Designed Garbage Collection"
- IPSJ Yamashita SIG Research Award, 2017 and 2018.
- Best Paper Award, IEEE BigComp, 2022 (2nd place) – "Accelerating Concurrency Control with Active Thread Adjustment"
- Best Paper Award, DSEA Workshop @ IEEE SNAMS, 2018 – "PPMDS: A Distributed Metadata Server Based on Nonblocking Transactions"
- Best Paper Award, IEEE BigComp, 2017 – "Accelerating Read Atomic Multi-Partition Transaction with Remote Direct Memory Access"

Service

Program Committee Member

- ACM International Conference on Management of Data (SIGMOD) (2027-)
- IEEE International Conference on Data Engineering (ICDE) (2027-)
- IEEE International Conference on Data Mining (ICDM) (2027-)
- International Conference on Web Search and Data Mining (WSDM) (2027-)
- International Conference on Very Large Data Bases (VLDB) (2026-)
- International Conference for High Performance Computing, Networking, Storage, and Analysis (SC) (2021)

Professional Memberships

- Association for Computing Machinery (ACM), Institute of Electrical and Electronics Engineers (IEEE), Information Processing Society of Japan (IPSJ), Institute of Electronics, Information and Communication Engineers (IEICE), Database Society of Japan (DBSJ)

Research Funding

- *A Foundation Model for Space Science from Massive Multimodal Data.* JST ARiSE, October 2026 – September 2031. Co-Principal Investigator.
- *Bespoke Transaction Processing System.* JST CRONOS, October 2026 – September 2031. Principal Investigator.
- *Hyperdimensional Photonic Tensor Core Utilizing Microcomb and Heterogeneous Integration.* JST CREST (Photonics Synergy), October 2025 – September 2030. Co-Investigator.
- *Foundations and Applications of Program Verification Techniques.* JSPS KAKENHI Grant-in-Aid for Scientific Research (S), April 2025 – March 2030. Co-Investigator.

- *Safe Database System.* SECOM Science and Technology Foundation, October 2024 – September 2028. Principal Investigator.
- *Creation of Agile Data Systems for Autonomous Mobile Robots.* JSPS KAKENHI Grant-in-Aid for Scientific Research (B), April 2022 – March 2025. Principal Investigator.
- *Creation of a Real-Time Data Kernel for Data-Intensive Science.* JSPS KAKENHI Grant-in-Aid for Scientific Research (B), April 2019 – March 2022. Principal Investigator.
- *Statistical Computational Cosmology with Big Astronomical Imaging Data.* JST CREST (Big Application), October 2014 – March 2020. Co-Investigator.
- *Research on Encrypted Data Processing Platform Systems.* JST A-STEP (Exploratory Type), 2014 – 2015. Principal Investigator.

Publications

Refereed Journal Articles

[1] Kodai Doki, Takashi Hoshino, and Hideyuki Kawashima. "Accelerating Scan Transactions on Scan-intensive Workloads with Node Locking". In: *Journal of Information Processing* 34 (2026), pp. 140–148.

[2] Tatsuhiro Nakamori and Hideyuki Kawashima. "Libra: One-Shot Parameter Sensitivity Estimation for Transfer Learning in Database Performance Prediction". In: *Proc. VLDB Endow.* 19.5 (2026), pp. 945–957.

[3] Rei Sato, Cui Gordon, Kazuhiro Saito, Hideyuki Kawashima, Tetsuro Nikuni and Shohei Watabe. "Two-Step Quantum Search Algorithm for Solving Traveling Salesman Problems". In: *IEEE Transactions on Quantum Engineering* vol. 6, pp. 1-12, 2025, Art no. 3100712.

[4] Aoi Kida and Hideyuki Kawashima. "Qsync: Making Simplified SMR Protocol Tolerant to Partial Network Partitioning". In: *IEICE Trans. Inf. & Syst.* E109-D.1 (Jan. 2026), pp. 13–22.

[5] Jun Nemoto, Takashi Kambayashi, Takashi Hoshino, and Hideyuki Kawashima. "Oze: Decentralized Graph-based Concurrency Control for Long-running Update Transactions". In: *Proc. VLDB Endow.* 18.8 (2025), pp. 2321–2333.

[6] Tung Nguyen and Hideyuki Kawashima. "Fairly Decentralizing a Hybrid Concurrency Control Protocol for Real-Time Database Systems". In: *Concurrency and Computation: Practice and Experience* 37.4–5 (2025).

[7] Aoi Kida and Hideyuki Kawashima. "Cataphract: A Batch Processing Method Specialized for BFT Databases". In: *International Journal of Networking and Computing* 15.1 (2025), pp. 32–50.

[8] Rina Onishi, Takashi Hoshino, and Hideyuki Kawashima. "EPO-R: An Efficient Garbage Collection Scheme for Long-Term Transactions". In: *Concurrency and Computation: Practice and Experience* 36.17 (2024).

[9] Masaru Uchida and Hideyuki Kawashima. "CLMD: Making Lock Manager Predictable and Concurrent for Deterministic Concurrency Control". In: *International Journal of Networking and Computing* 14.1 (2024), pp. 81–92.

[10] Masahide Fukuyama, Masahiro Tanaka, Ryota Ogino, and Hideyuki Kawashima. "eSilo: Making Silo Secure with SGX". In: *International Journal of Networking and Computing* 14.2 (2024), pp. 206–224.

[11] Yushi Ogiwara, Ayanori Yorozu, Akihisa Ohya, and Hideyuki Kawashima. "Transactional TF: Transform Library with Concurrency and Correctness". In: *IEICE Trans. Inf. & Syst.* E106-D.12 (2023), pp. 1951–1959.

[12] Tatsuhiro Nakamori, Jun Nemoto, Takashi Hoshino, and Hideyuki Kawashima. “Scalable Timestamp Allocation for Deadlock Prevention in Two-phase Locking based Protocols”. In: *Journal of Information Processing* 31 (2023), pp. 365–374.

[13] Natsuki Hamada, Kazuhiro Saito, and Hideyuki Kawashima. “Applying Quantum Annealing for Shift Scheduling Problem for Call Centers”. In: *International Journal of Networking and Computing* 13.1 (2023), pp. 2–17.

[14] Kosei Masumura, Takashi Hoshino, and Hideyuki Kawashima. “Fast Concurrency Control with Thread Activity Management beyond Backoff”. In: *Journal of Information Processing* 30 (2022), pp. 552–561.

[15] Takayuki Tanabe, Takashi Hoshino, Hideyuki Kawashima, and Osamu Tatebe. “An Analysis of Concurrency Control Protocols for In-Memory Databases with CCBench”. In: *Proc. VLDB Endow.* 13.13 (2020), pp. 3531–3544.

[16] Kohei Hiraga, Osamu Tatebe, and Hideyuki Kawashima. “Scalable Distributed Metadata Server Based on Nonblocking Transactions”. In: *Journal of Universal Computer Science* 26.1 (2020), pp. 89–106.

[17] Yasuhiro Nakamura, Hideyuki Kawashima, and Osamu Tatebe. “Integration of TicToc Concurrency Control Protocol with Parallel Write Ahead Logging Protocol”. In: *International Journal of Networking and Computing* 9.2 (2019), pp. 339–353.

[18] Makoto Yabuta, Anh Nguyen, Shinpei Kato, Masato Edahiro, and Hideyuki Kawashima. “Relational Joins on GPUs: A Closer Look”. In: *IEEE Transactions on Parallel and Distributed Systems* 28.9 (Sept. 2017), pp. 2663–2673.

[19] Li Jiang, Hideyuki Kawashima, and Osamu Tatebe. “Efficient Window Aggregate Method on Array Database System”. In: *Journal of Information Processing* 24.6 (2016), pp. 867–877.

[20] Yasin Oge, Masato Yoshimi, Takefumi Miyoshi, Hideyuki Kawashima, Hidetsugu Irie, and Tsutomu Yoshinaga. “Design and Evaluation of a Configurable Query Processing Hardware for Data Streams”. In: *IEICE Trans. Inf. & Syst.* E98-D.12 (2015), pp. 2207–2217.

[21] Hideyuki Kawashima. “Data Processing Technologies for Realizing Intelligent Environments”. In: *IEICE Journal* 97.8 (Aug. 2014), pp. 675–678. In Japanese.

Refereed Conference & Workshop Papers

[1] Tatsuhiro Nakamori, Masahide Fukuyama, Yoshihiko Suhara, and Hideyuki Kawashima. “The Occupational Impact of Cybernetic Avatars on the Japanese Labor Market”. In: *Proceedings of the ICRES 2025 Conference*. Lecture Notes in Networks and Systems 1556. Springer, 2026, pp. 219–230.

[2] Aoi Kida and Hideyuki Kawashima. “Qsync: Extending Simplified SMR Protocol with Partial Network Partition Tolerance”. In: *Proceedings of the International Conference on High Performance Computing in Asia-Pacific Region*. HPC Asia ’25. 2025, pp. 120–130.

[3] Masaru Uchida and Hideyuki Kawashima. “Campus: Making Cluster-based Vector Index Scalable with Transactions”. In: *IEEE International Conference on Data Engineering Workshops*. ICDEW ’25. 2025, pp. 115–121.

[4] Kodai Doki, Takashi Hoshino, and Hideyuki Kawashima. “WoundDie: Concurrency Control Protocol with Lightweight Priority Control”. In: *Proceedings of the 15th ACM SIGOPS Asia-Pacific Workshop on Systems*. APSys ’24. 2024, pp. 130–135.

[5] Haowen Li, Rina Onishi, and Hideyuki Kawashima. “Serval: A Wait-free Multi-version Deterministic Concurrency Control Scheme”. In: *International Symposium on Computing and Networking*.

CANDAR '24. 2024, pp. 169–175.

[6] Yutaro Bessho and Hideyuki Kawashima. "Fast Hybrid Transactional Index with Scalable Phantom Avoidance". In: *CANDAR Workshops*. 2024, pp. 36–42.

[7] Hideyuki Kawashima. "Hippo: Accelerating Transaction Processing for Approximate Query Processing Engine with Sampling Semantics". In: *CANDAR Workshops*. 2024, pp. 117–122.

[8] Yushi Ogiwara and Hideyuki Kawashima. "Sparse Ternary Matrix Multiplication with Tensor Core for Transformer". In: *CANDAR Workshops*. 2024, pp. 150–156.

[9] Shuma Ohara, Kazuhiro Saito, and Hideyuki Kawashima. "Evaluation of QAOA for Black-Box Serializability Problem". In: *CANDAR Workshops*. 2024, pp. 190–196.

[10] Gordon Cui, Rei Sato, Kazuhiro Saito, Rodney Van Meter, and Hideyuki Kawashima. "Hybrid Quantum Search Algorithm for Solving the Multi-Dimensional Knapsack Problem". In: *IEEE International Conference on Quantum Computing and Engineering*. QCE '24. Sept. 2024, pp. 502–503.

[11] Natsuki Hamada, Kazuhiro Saito, and Hideyuki Kawashima. "Qobra: Fast Verification of Transactional Serializability with Quantum Annealing". In: *IEEE International Conference on Big Data*. BigData '23. Dec. 2023, pp. 1322–1329.

[12] Tatsuhiro Nakamori, Jun Nemoto, Takashi Hoshino, and Hideyuki Kawashima. "Removing Performance Bottleneck of Timestamp Allocation in Two-Phase Locking Based Protocol". In: *Australasian Database Conference*. ADC '22. Lecture Notes in Computer Science 13459. 2022, pp. 201–208.

[13] Tatsuhiro Nakamori, Jun Nemoto, Takashi Hoshino, and Hideyuki Kawashima. "Decentralization of Two Phase Locking based Protocols". In: *Proceedings of the 31st International Symposium on High-Performance Parallel and Distributed Computing*. HPDC '22. June 2022, pp. 281–282.

[14] Kosei Masumura, Takashi Hoshino, and Hideyuki Kawashima. "Accelerating Concurrency Control with Active Thread Adjustment". In: *IEEE International Conference on Big Data and Smart Computing*. BigComp '22. 2022, pp. 280–287.

[15] Masahiro Tanaka and Hideyuki Kawashima. "Stable Low Latency Logging for Epoch-based In-memory Database". In: *IEEE International Conference on Big Data and Smart Computing*. BigComp '22. 2022, pp. 167–170.

[16] Rina Onishi, Takashi Hoshino, and Hideyuki Kawashima. "EPO-R: An Efficient Garbage Collection Scheme for Long-Term Transactions". In: *International Symposium on Computing and Networking*. CANDAR '22. 2022, pp. 144–150.

[17] Yushi Ogiwara, Ayanori Yorozu, Akihisa Ohya, and Hideyuki Kawashima. "Transactional Transform Library for ROS". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems*. IROS '22. 2022, pp. 9722–9727.

[18] Yushi Ogiwara, Ayanori Yorozu, Akihisa Ohya, and Hideyuki Kawashima. "Making ROS TF Transactional". In: *ACM/IEEE International Conference on Cyber-Physical Systems*. ICCPS '22. 2022, pp. 318–319.

[19] Natsuki Hamada, Kazuhiro Saito, and Hideyuki Kawashima. "Practical Effectiveness of Quantum Annealing for Shift Scheduling Problem". In: *IEEE International Parallel and Distributed Processing Symposium Workshops*. IPDPSW '22. May 2022, pp. 421–424.

[20] Mengfei Shen, Hideyuki Kawashima, and Kazuhiro Saito. "Fast Accurate Discovery of Tuple Inclusion Dependencies". In: *IEEE International Conference on Smart Computing*. SMARTCOMP '22. 2022, pp. 246–251.

[21] Takuto Ogura, Yoshiki Akita, Yuki Miyazawa, and Hideyuki Kawashima. "Accelerating Geo-

Distributed Transaction Processing with Fast Logging". In: *IEEE International Conference on Big Data*. BigData '21. 2021, pp. 2390–2399.

[22] Harunobu Daikoku, Hideyuki Kawashima, and Osamu Tatebe. "Skew-Aware Collective Communication for MapReduce Shuffling". In: *IEEE International Conference on Big Data*. BigData '18. 2018, pp. 3331–3340.

[23] Masahiro Tanaka, Osamu Tatebe, and Hideyuki Kawashima. "Applying Pwrake Workflow System and Gfarm File System to Telescope Data Processing". In: *IEEE International Conference on Cluster Computing*. CLUSTER '18. 2018, pp. 124–133.

[24] Yasuhiro Nakamura, Hideyuki Kawashima, and Osamu Tatebe. "Integrating TicToc with Parallel Logging". In: *CANDAR Workshops*. 2018, pp. 105–111.

[25] Kohei Hiraga, Osamu Tatebe, and Hideyuki Kawashima. "PPMDS: A Distributed Metadata Server Based on Nonblocking Transactions". In: *International Conference on Social Networks Analysis, Management and Security*. SNAMS '18. 2018, pp. 202–208.

[26] Takayuki Tanabe, Hideyuki Kawashima, and Osamu Tatebe. "Integration of Parallel Write Ahead Logging and Cicada Concurrency Control Method". In: *IEEE International Conference on Smart Computing*. SMARTCOMP '18. 2018, pp. 291–296.

[27] Kentaro Horio, Hideyuki Kawashima, and Osamu Tatebe. "Efficient Parallel Summation on Encrypted Database System". In: *IEEE International Conference on Big Data and Smart Computing*. BigComp '17. 2017, pp. 178–185.

[28] Naofumi Murata, Hideyuki Kawashima, and Osamu Tatebe. "Accelerating Read Atomic Multi-Partition Transaction with Remote Direct Memory Access". In: *IEEE International Conference on Big Data and Smart Computing*. BigComp '17. 2017, pp. 239–246.

[29] Ryuya Mitsuhashi, Hideyuki Kawashima, Takahiro Nishimichi, and Osamu Tatebe. "Three-Dimensional Spatial Join Count Exploiting CPU Optimized STR R-Tree". In: *IEEE International Conference on Big Data*. BigData '16. 2016, pp. 2938–2947.

[30] Li Jiang, Hideyuki Kawashima, and Osamu Tatebe. "Fast Window Aggregate on Array Database by Recursive Incremental Computation". In: *IEEE International Conference on e-Science*. eScience '16. 2016, pp. 101–110.

[31] Harunobu Daikoku, Hideyuki Kawashima, and Osamu Tatebe. "On Exploring Efficient Shuffle Design for In-Memory MapReduce". In: *BeyondMR Workshop @ SIGMOD*. 2016, 6:1–6:10.

[32] Yasin Oge, Takefumi Miyoshi, Hideyuki Kawashima, and Tsutomu Yoshinaga. "A Fast Handshake Join Implementation on FPGA with Adaptive Merging Network". In: *International Conference on Scientific and Statistical Database Management*. SSDBM '13. 2013.

[33] Yasin Oge, Masato Yoshimi, Takefumi Miyoshi, Hideyuki Kawashima, Hidetsugu Irie, and Tsutomu Yoshinaga. "An Efficient and Scalable Implementation of Sliding-Window Aggregate Operator on FPGA". In: *International Symposium on Computing and Networking*. CANDAR '13. 2013, pp. 112–121.

[34] Takashi Takagi, Hideyuki Kawashima, Toshiyuki Amagasa, and Hiroyuki Kitagawa. "Providing Constructed Buildings Information by ASTER Satellite DEM Images and Web Contents". In: *International Conference on Database Systems for Advanced Applications*. DASFAA '10. 2010.

[35] Satoru Satake, Hideyuki Kawashima, Michita Imai, Kenshiro Hirose, and Yuichiro Anzai. "IRIOS: Interactive News Announcer Robot System". In: *APWeb Workshops*. 2006, pp. 733–740.

[36] Hideyuki Kawashima, Yutaka Hirota, Satoru Satake, and Michita Imai. "MeT: A Real World Oriented

Metadata Management System for Semantic Sensor Networks". In: *Workshop on Data Management for Sensor Networks*. DMSN '06. 2006, pp. 13–18.

[37] Michita Imai, Yutaka Hirota, Satoru Satake, and Hideyuki Kawashima. "Semantic Sensor Network for Physically Grounded Applications". In: *International Conference on Control, Automation, Robotics and Vision*. ICARCV '06. 2006, pp. 1–6.

[38] Kenshiro Hirose, Satoru Satake, Hideyuki Kawashima, Michita Imai, and Yuichiro Anzai. "Development of Communication Contents Description Language". In: *IEEE International Conference on Systems, Man and Cybernetics*. SMC '04. 2004, pp. 2896–2900.

[39] Hideyuki Kawashima, Motomichi Toyama, Michita Imai, and Yuichiro Anzai. "Providing Persistence for Sensor Stream with Temporal Consistency Conscious WAL". In: *International Conference on Information Systems and Databases*. ISDB '02. 2002, pp. 13–18.

[40] Hideyuki Kawashima, Michita Imai, Motomichi Toyama, and Yuichiro Anzai. "Providing Persistence for Sensor Streams with Light Neighbor WAL". In: *IEEE Pacific Rim International Symposium on Dependable Computing*. PRDC '02. 2002, pp. 257–264.

Book Chapters & Edited Volumes

[1] Hiroyuki Kitagawa, Yousuke Watanabe, Hideyuki Kawashima, and Toshiyuki Amagasa. *Stream-based Real World Information Integration Framework*. Springer, Dec. 2010.

Preprints

[1] Tatsuhiro Nakamori, Laura Gomezjurado Gonzalez, Ganesh Talluri, Ansh Tiwari, Hideyuki Kawashima, Ioannis Mitliagkas, Guillaume Rabusseau, and Hiroki Naganuma. "Orth-Dion: Eliminating Geometric Mismatch in Distributed Low-Rank Spectral Optimization". In: *CoRR* abs/2605.16341 (2026).

[2] Takayuki Tanabe, Shinichi Umegane, Suguru Arakawa, Ryoji Kurosawa, Takashi Hoshino, Hideyuki Kawashima, Masahiro Tanaka, and Takashi Kambayashi. "Shirakami: A Hybrid Concurrency Control Protocol for Tsurugi Relational Database System". In: *CoRR* abs/2303.18142 (2026).

[3] Jun Nemoto, Takashi Kambayashi, Takashi Hoshino, and Hideyuki Kawashima. "Oze: Decentralized Graph-based Concurrency Control for Long-running Update Transactions (Extended Version)". In: *CoRR* abs/2210.04179 (2025).

[4] Sho Nakazono, Yutaro Bessho, Hideyuki Kawashima, and Tatsuhiro Nakamori. "Griffin: Fast Transactional Database Index with Hash and B+-Tree". In: *CoRR* abs/2407.13294 (2024).

[5] Sho Nakazono, Hiroyuki Uchiyama, Yasuhiro Fujiwara, and Hideyuki Kawashima. "Scheduling Space Expander: An Extension of Concurrency Control for Data Ingestion Queries". In: *CoRR* abs/2301.10440 (2023).